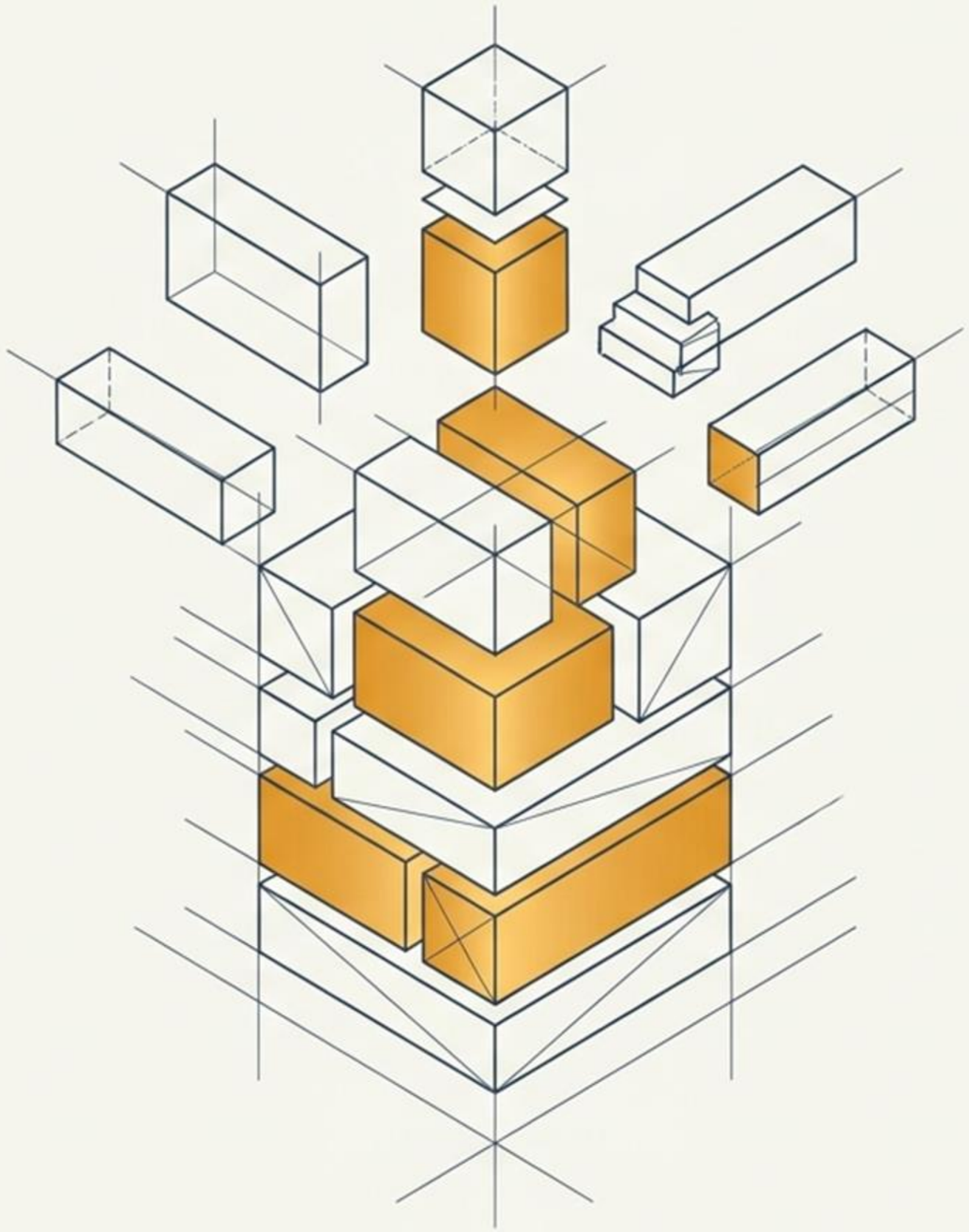
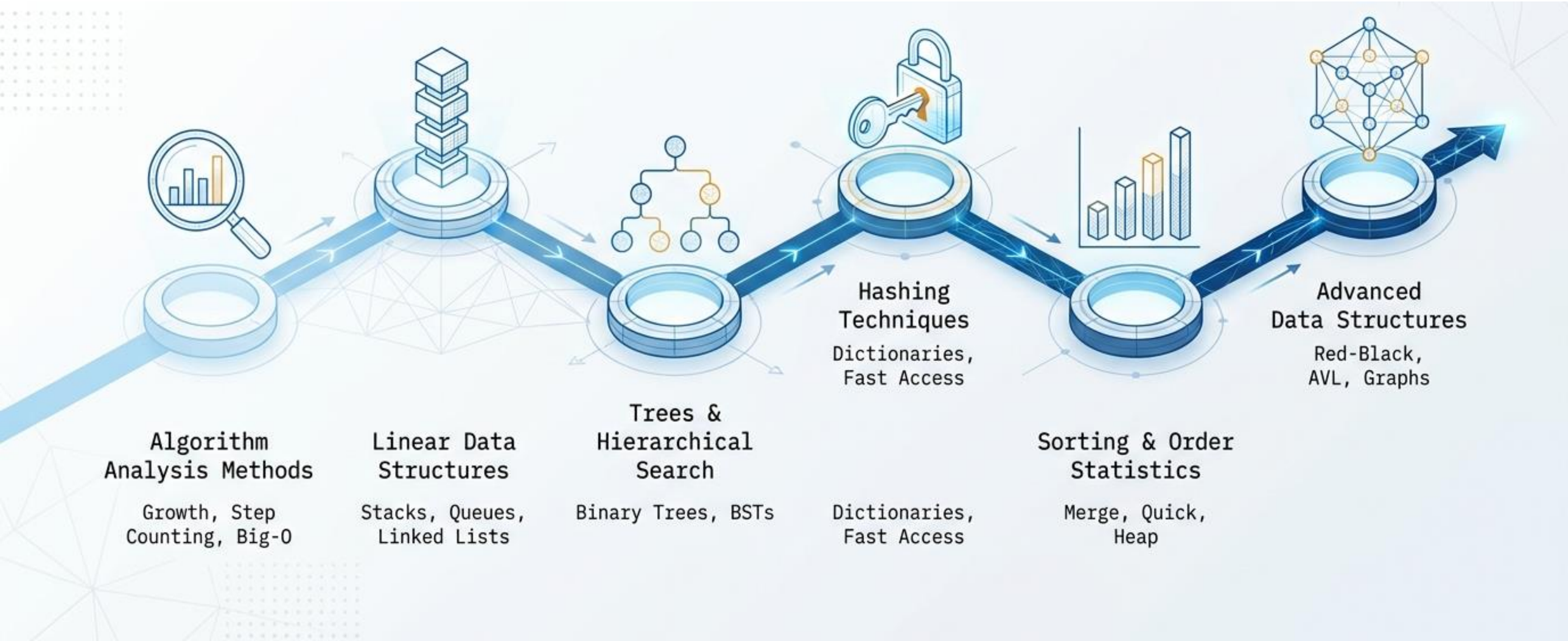


Introduction to Data Structures

Course Roadmap



Evaluation Breakdown

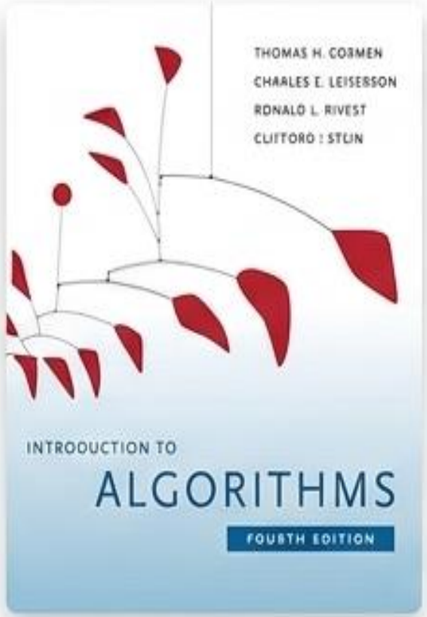
60% — Final Examination (Comprehensive)

20% — Midterm Examination

10% — Class Participation & Quizzes

10% — Homework Exercises & Project

Required & Recommended Texts



PRIMARY TEXTBOOK

Introduction to Algorithms

4th Edition (2022) | MIT Press

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein

Supplemental Reading

- Horowitz & Sahni — Fundamentals of Data Structures
- Mohammad Ghodsi — Data Structures and Fundamentals of Algorithms (Fatemi Publishing)

Contact & Office Location



Email

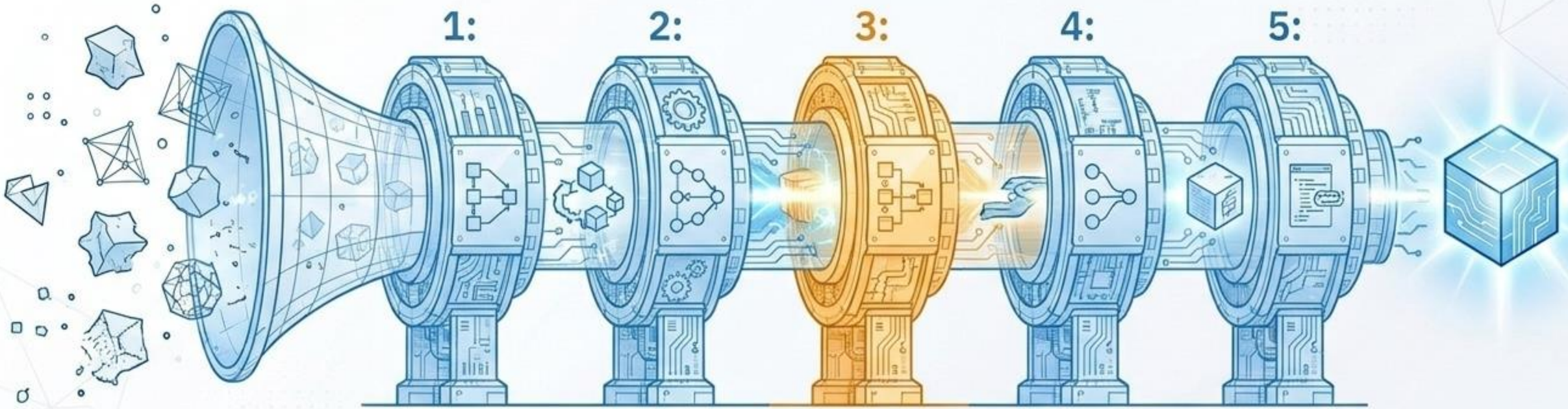
eghdami.mahdie@mail.um.ac.ir



Location

First Floor, Room 716

Engineering a solution requires moving through a **five-stage computational pipeline.**



1: Abstraction (Modeling)

2: Algorithm (Logic)

3: Data Structure (Organization)

4: Analysis (Evaluation)

5: Program (Execution)

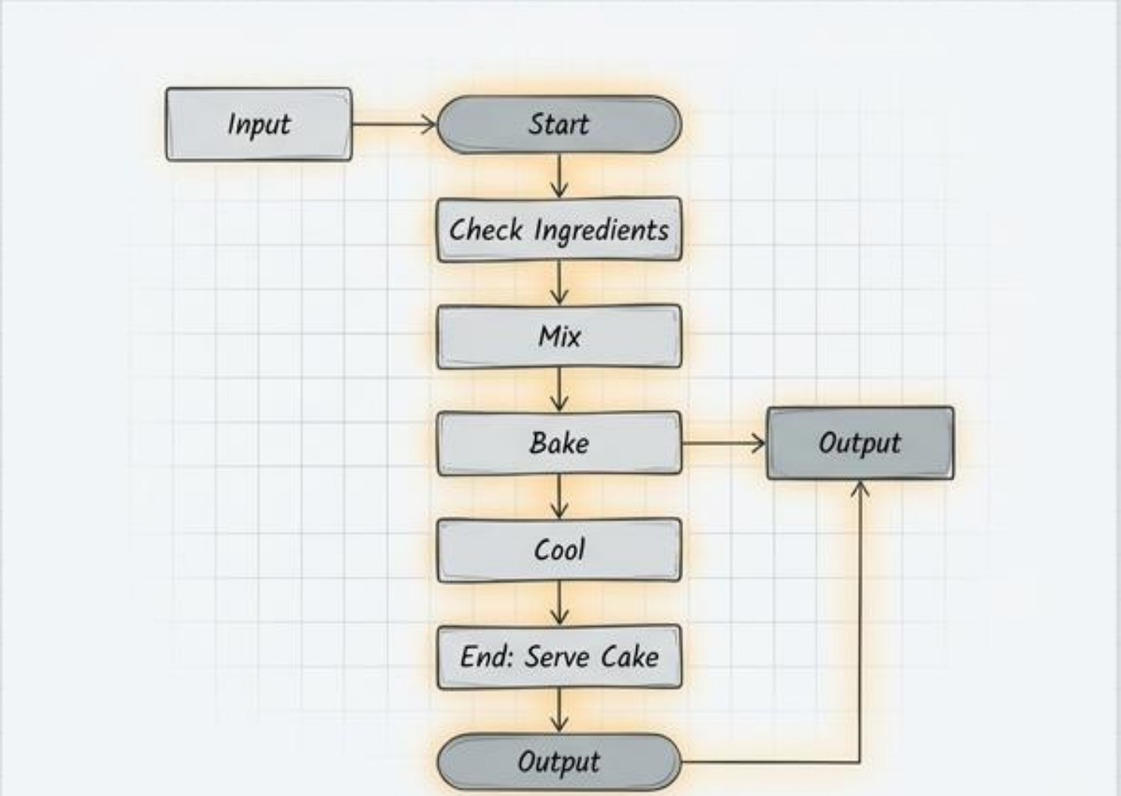
The Data Storage Container
(Data Structure)

How we organize, store, and hold our information.

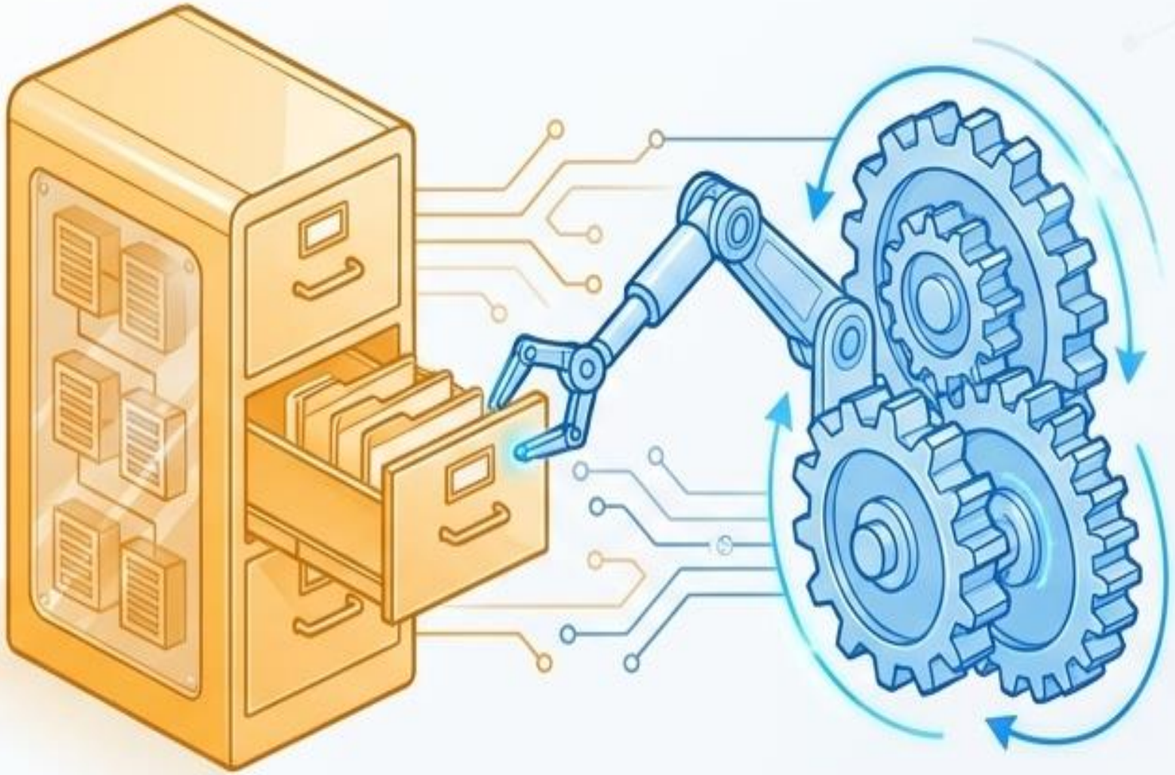


The Step-by-Step Recipe
(Algorithm)

What we do with that information to solve a problem.



Computation relies on the perfect pairing of **structural storage** and **active transformation.**



Data Structure

How we organize, store, and encapsulate data in memory (e.g., A Phonebook)

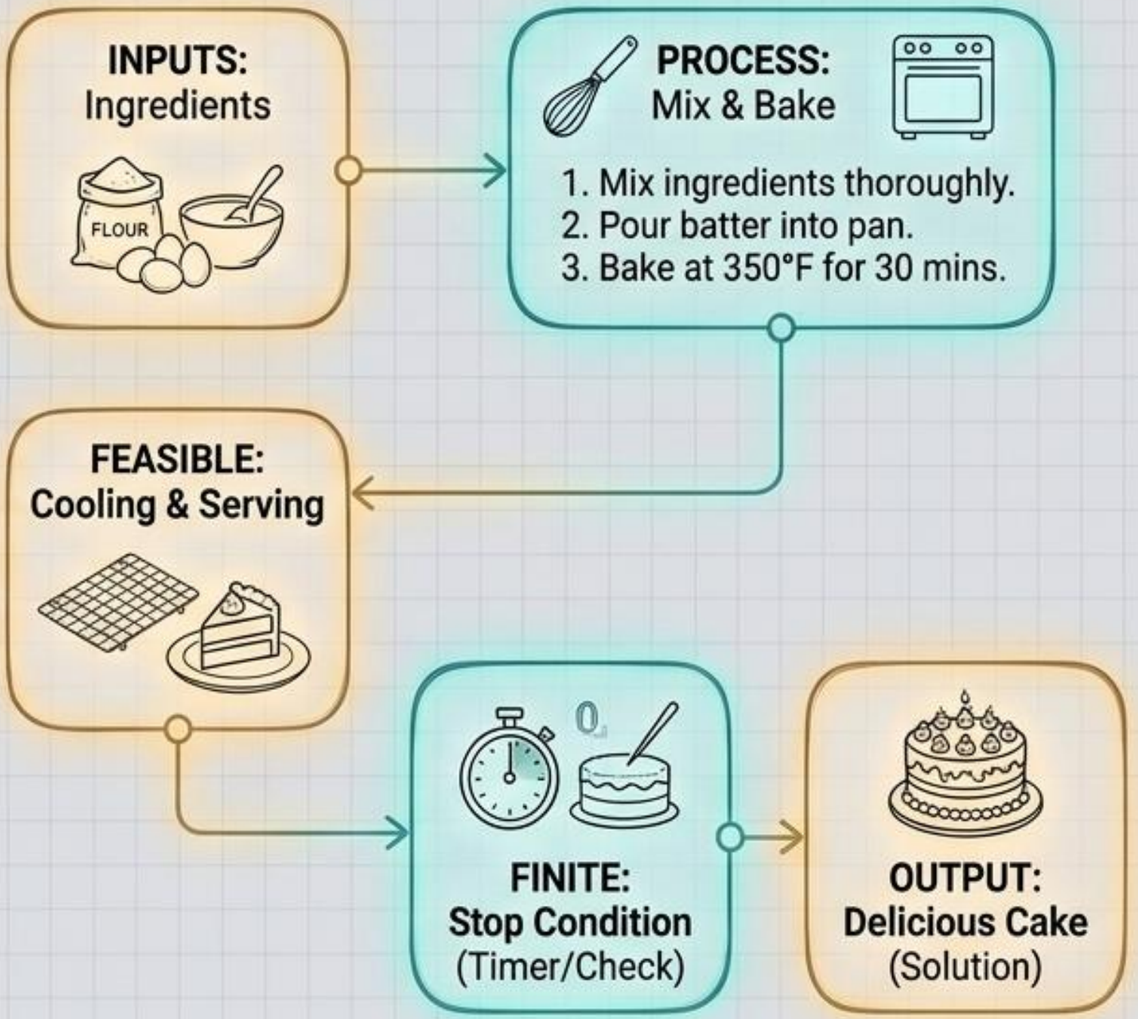
Algorithm

The sequence of computational steps that transform input into output (e.g., Searching for a name).

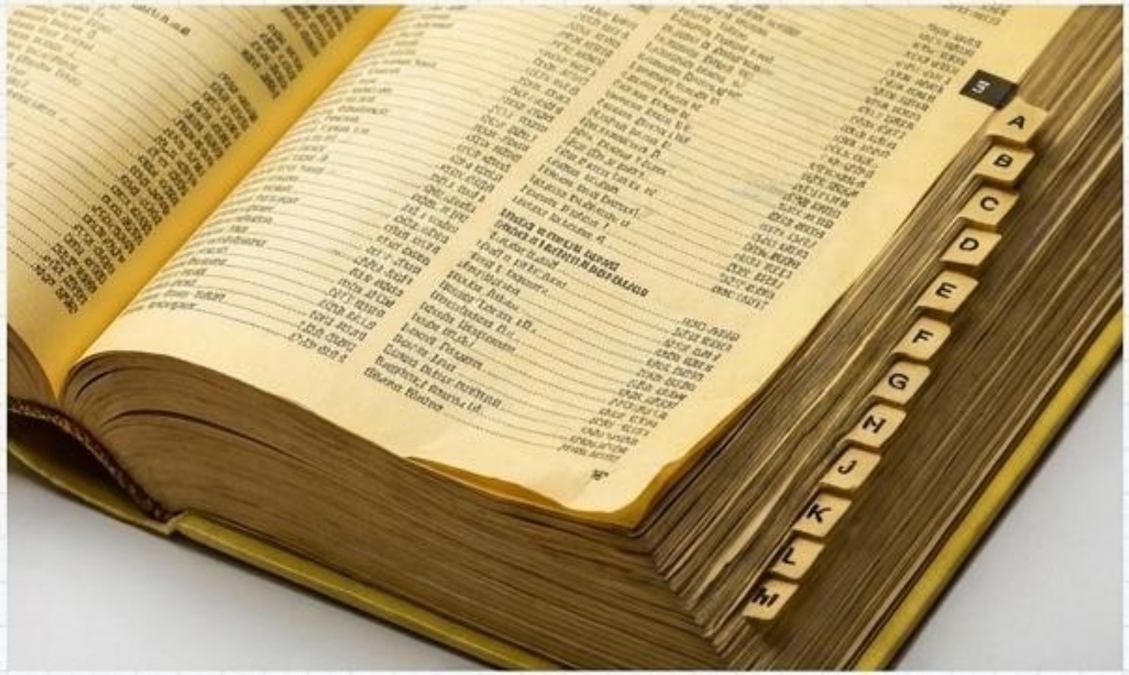
Anatomy of a Good Recipe

To be considered a true algorithm, a process must check these five boxes:

- ☑ **1. Gets Inputs**
  (Starts with ingredients)
- ☑ **2. Produces Output**
  (Results in a solution)
- ☑ **3. Unambiguous**
  (Every single step is perfectly clear)
- ☑ **4. Finite**
  (It cannot run forever; it must stop)
- ☑ **5. Feasible**
  (It can actually be done in the real world)



Why Organization Matters



The Phonebook

Organized alphabetically. Perfect for fast searching when you know exactly who you are looking for.



The Hospital ER

Organized by priority and severity. First-come, first-serve would be a disaster here.

Key Takeaway: The way we store data changes how efficiently we can use it.

Measuring Success

How do we know if our container and recipe are actually good?
We measure two limited resources:



1. Time (Speed)

How long the process takes as the amount of data grows.

2. Space (Memory)

How much storage capacity the process eats up while running.

The Hospital ER Example

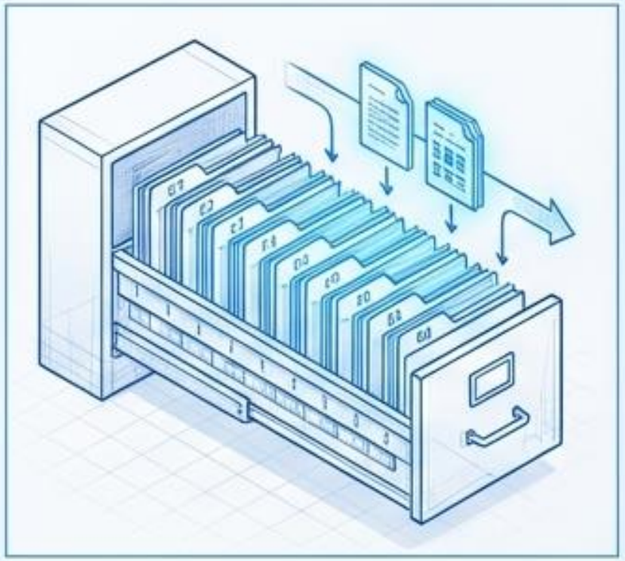
Unsorted Array



Insert is fast: **O(1)**

Finding the critical patient is
Finding the critical patient is dangerously slow: **O(n)** (requires checking every single person).

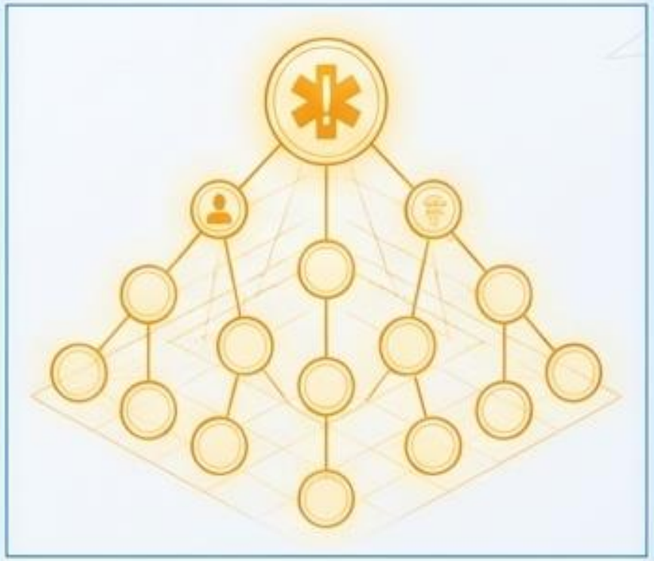
Sorted Array



Finding the critical patient is instant: **O(1)**

Inserting a new arrival is terribly slow: **O(n)** (requires shifting everyone else down).

Max-Heap Priority Queue



The 'Goldilocks' ADT.

Both Inserts and Searches are fast: **O(log n)**